Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

Application: Data Scientist

Dear Hiring Team,

I am applying for the Data Scientist position. I am a mechatronics and robotics engineer doing my PhD at Aalborg University, with my thesis expected to be finished in March. My work has always been close to real data from real systems: collecting it carefully, cleaning it, and using it to make decisions about a design.

In my PhD I collected EMG and motion data from 12 participants for a shoulder exoskeleton study, using the Qualisys data acquisition system and motion capture cameras. For a second study with 6 participants, I implemented an Arduino-based trigger system to synchronise a robot's motion with the Qualisys recording, so data from different systems lined up in time. I designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance. I also optimised my exoskeleton design with a genetic algorithm.

In my Master's I fed experimental data from a pneumatic artificial muscle into a neural network developed by a colleague, and used its output to design the outer PID loop of a gain-scheduled controller in MATLAB/Simulink, reaching 0.3 to 0.78% position error (IEEE CONECCT 2022). That taught me how much a model depends on the quality of the data behind it.

I want to be honest that my main tool is MATLAB, and my Python is at a basic level. I have not yet built machine learning models from scratch in industry. What I bring is a solid base in measurement, modelling and optimisation, and a careful way of working with data. I learn quickly, and I would like to grow my Python and machine learning skills in your team. I also present results clearly, from teaching university courses and writing papers.

Thank you for considering my application. You can see my work at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder